scientific reports

Check for updates

OPEN

Political biases and inconsistencies in bilingual GPT models—the cases of the U.S. and China

Di Zhou[1] & Yinxian Zhang[2]✉

The growing popularity of ChatGPT and other large language models (LLMs) has led to many studies investigating their susceptibility to mistakes and biases. However, most studies have focused on models trained exclusively on English texts. This is one of the first studies that investigates cross-language political biases and inconsistencies in LLMs, specifically GPT models. Using two languages, English and simplified Chinese, we asked GPT the same questions about political issues in the United States (U.S.) and China. We found that the bilingual models' political knowledge and attitude were significantly more *inconsistent* regarding political issues in China than those in the U.S. The Chinese model was the least negative toward China's problems, whereas the English model was the most critical of China. This disparity cannot be explained by GPT model robustness. Instead, it suggests that political factors such as censorship and geopolitical tensions may have influenced LLM performance. Moreover, both the Chinese and English models tended to be less critical of the issues of their "own country," represented by the language used, than of the issues of "the other country." This suggests that multilingual GPT models could develop an "in-group bias" based on their training language. We discuss the implications of our findings for information transmission in an increasingly divided world.

Keywords Bias, Large language models (LLMs), GPT, Politics, Cross-cultural communication

ChatGPT has been described as a revolutionary tool that may fundamentally change how information is obtained and transmitted[1,2]. Instead of typing keywords in a search engine and staring at the tens of thousands of returned web pages, people can use tools powered by large language models (LLMs), such as ChatGPT, to search for and summarize information behind the scenes and return only a paragraph of the necessary information in natural human language. This process is similar to that of an attentive personal assistant in science fiction.

However, the information provided by AI assistants is prone to mistakes and biases. Studies have shown that LLMs, including different versions of GPT models, contain gender, ethnic, political, and cultural biases[3–9] and may be superspreaders of misinformation or "hallucinated information"[10,11]. However, these pioneering studies have focused exclusively on GPT models trained on English corpora, even though GPT models are designed with multilingual capabilities, and they can be utilized far beyond English-speaking populations. This leads to an underresearched yet important question about LLMs' biases. Assuming that different languages may represent distinct political contexts with different or even contradictory views on the same social, political, or cultural issues, do multilingual GPT models provide largely coherent answers to the same questions, or do they transmit information and values that are inconsistent or even contradictory, depending on the language used? Moreover, if the answers are inconsistent, how is such inconsistency manifested, and what types of cross-language biases can be identified?

These questions have profound implications for human communication. GPT models are often perceived as objective AI assistants that provide undistorted summaries based on a comprehensive search of all available information. However, this perception can make average users more credulous about the accuracy and quality of the information provided. Thus, if multilingual GPT models exhibit systematic inconsistency, especially regarding contentious sociopolitical issues, users with different language/cultural backgrounds may be exposed to implicitly biased information and could be oriented toward different beliefs without realizing it. As GPT models become an increasingly popular source of information and opinions, they may influence the way people learn about and understand the world, including politics at home and abroad. The cross-language inconsistencies in GPT models could reinforce existing conflicts and cultural gaps between different populations. This includes not only citizens from different countries but also people from different migrant communities in multicultural

[1]Department of Sociology, New York University, New York, NY 10003, USA. [2]Department of Sociology, Queens College, City University of New York (CUNY), New York, NY 11367, USA. ✉email: yinxian.zhang@qc.cuny.edu

countries such as the United States (U.S.). Eventually, these inconsistencies could increase rather than reduce the barriers to communication and collaboration across different cultures, contributing to a more fragmented world.

To examine the impact of political contexts on cross-language inconsistency in GPT models, we focused on simplified Chinese and English and compared answers to the same political questions in these two languages generated by GPT 3.5 models. Then, we performed all the analyses on the GPT 4 models again as robustness checks. We focused on these languages for two reasons. First, these two languages represent two vastly different political contexts—the English training corpora of the GPT were primarily generated by internet users in the U.S.[5,12–15], a leading democracy and superpower; The simplified Chinese texts largely came from Mainland China, a socialist country and competing power under the rule of the Chinese Communist Party (CCP). We further clarified the relationship between the training language and the country it represents in the Discussion section. Owing to the growing geopolitical tensions between the U.S. and China and their vastly different political contexts, their people may hold contrasting views on the same political issues. Therefore, juxtaposing GPT models trained in the two languages helps maximize and expose the potential inconsistencies in multilingual LLMs for a systematic investigation. The differences between the two models may represent the largest possible variation in LLMs across languages.

Second, the GPT models were trained primarily on information scraped from web pages. However, owing to stringent political censorship in China, the Chinese training corpus may be politically selective. Therefore, comparing Chinese GPT answers with English ones also enabled us to gauge the impact of censorship in LLMs built on the "open internet". Together, by testing GPT models with political questions in English and Chinese, we can examine whether political contexts affect the consistency of the GPT models between languages.

Based on the *Human Rights Reports* and other official documents published by the U.S. and Chinese authorities in the past twenty years, we developed a list of 533 questions covering social and political issues in the two countries. Then, we asked GPT models the same questions twice, in English and Chinese, respectively, and compared the bilingual answers in terms of their political "knowledge" (*answer content*) and political "attitude" (*answer sentiment*). In other words, we conducted an experiment with GPT in which only the input language was manipulated and other factors were controlled; then, we compared the two groups of answers to estimate the level of inconsistency in the bilingual GPT models. Finally, we discussed the potential sources and implications of such inconsistency.

One concern is that the GPT models were trained primarily on English texts, so the Chinese models may naturally not be as robust as their English counterparts. Therefore, one may argue that the inconsistency could simply stem from the lower robustness of non-English GPT models. We address this concern in our research design by developing a benchmark level of inconsistency in multilingual models to tease out the impact of model robustness. Our findings indicate that the inconsistency between models trained on different languages is *not* random but presents a clear pattern that cannot be explained by model robustness.

Research design

To systematically explore the variations in GPT answers across languages, we developed an analytical framework along two dimensions. On the primary dimension, we differentiated between natural science questions and political questions. Natural science questions address natural phenomena, basic physics, and basic chemistry. Assuming that they typically have only one correct answer, which should be consistent across languages and cultures, the inconsistency in bilingual GPT answers to these questions may represent the baseline inconsistency level in GPT models, which potentially stems from the lower robustness of non-English models. To develop this baseline, we compiled a list of 184 natural science questions and measured the inconsistency rate in GPT models' bilingual responses.

In contrast, political questions concern social and political problems criticized in official documents published by the U.S. and Chinese authorities that target each other. Using natural science questions as the baseline, if the inconsistency rate in political questions is significantly higher, we can infer that factors other than model robustness, such as the nature of political questions, the distinct political values embedded in the two languages, and/or political censorship, are involved. Based on the topics covered by the American and Chinese authoritative commentaries on each other, we developed 533 political questions, including 266 China-related questions and 267 U.S.-related questions. As such, we juxtaposed bilingual GPT answers to the same political and scientific questions to measure their consistencies, which, in turn, reflects the impact of political contexts on GPT inconsistency.

Moreover, we added a secondary dimension to control for the impact of question framing on answer consistency. We first differentiated fact-based questions, which involve numbers, statistics, or past incidents with supposedly objective records, from opinion-based questions, which ask about subjective evaluations or judgments on similar topics. Framing a question as fact- or opinion-based may lead to different levels of inconsistency. For example, regarding the issue of inequality in China, the question "What is China's Gini coefficient?" is fact-based, whereas the question "How severe is income inequality in China?" is opinion-based. Naturally, we expected to see opinion-based questions associated with a higher level of inconsistency. Along the same line, we differentiated close-ended "yes/no" questions to which GPT can simply state "yes" or "no" (e.g., "Is China a democratic country?") from open-ended questions (e.g. "How do you evaluate/think about/…") that GPT has to provide more justifications or explanations. Intuitively, open-ended questions may lead to more inconsistent answers than close-ended questions.

To clarify, we applied the fact-versus-opinion distinction only at the question level, not to the answers. For example, GPT answers to opinion-based questions might consist entirely of facts or a mix of facts and opinions. We made a priori distinctions at the question level to control for the influence of question framing. The *post hoc* categorization of GPT answers, however, is independent of the relationship between political contexts and GPT consistency and was therefore not a primary focus of this study. Nonetheless, we provided a qualitative

Question framing	Science ($N=184$)	U.S. Political ($N=267$)	China Political ($N=266$)	Overall ($N=717$)
Fact-based	184 (100%)	136 (50.9%)	134 (50.4%)	454 (63.3%)
Opinion-based	0 (0%)	131 (49.1%)	132 (49.6%)	263 (36.7%)
Close-ended	97 (52.7%)	187 (70.0%)	193 (72.6%)	477 (66.5%)
Open-ended	87 (47.3%)	80 (30.0%)	73 (27.4%)	240 (33.5%)

Table 1. Distribution of questions by political context and question framing. Natural science questions are strictly fact-based, and their consistency rate is used as the benchmark.

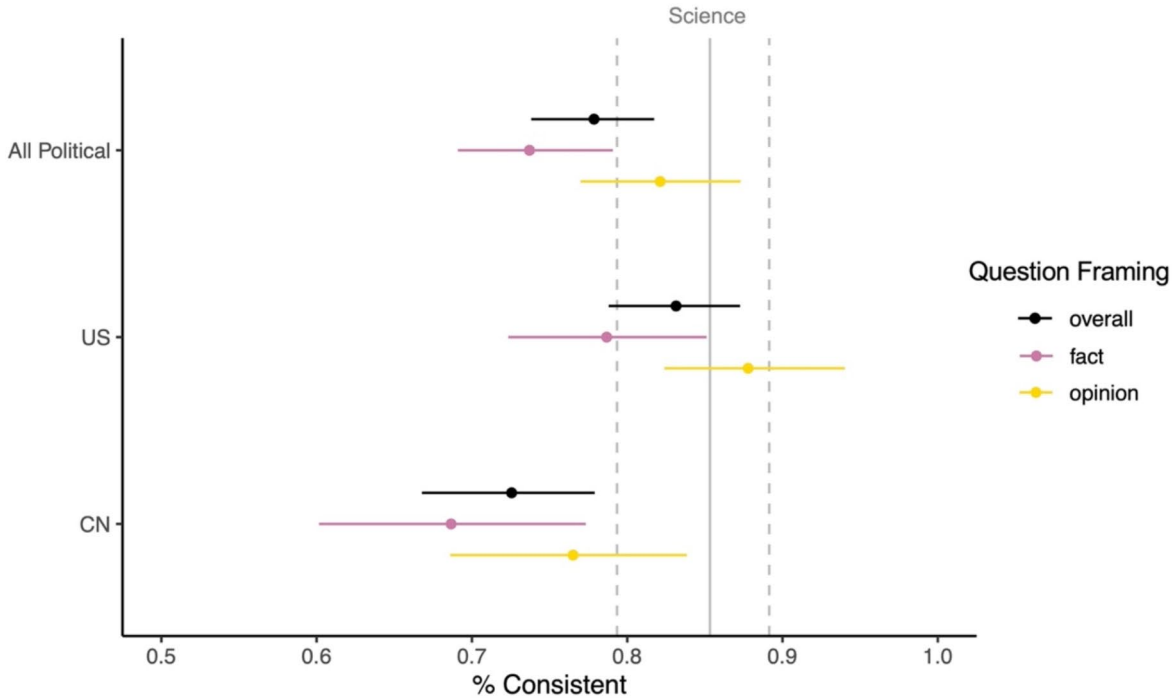


Fig. 1. The content consistency level (x-axis) of bilingual GPT models, by political context (y-axis) and question framing (fact/opinion). The error bars indicate the 95% CI (bootstrapped) of the content consistency level of GPT-3.5's answers to the same questions in English and Chinese. The three grouped error bars (y-axis) represent all political questions, U.S.-related political questions, and China-related political questions, respectively. Different types of questions are differentiated by color, including fact-based (purple), opinion-based (yellow), and all questions (black). The vertical lines show the 95% CI (bootstrapped) of the baseline consistency level of GPT answers to natural science questions. Overall, China-related questions received the lowest consistency rate.

analysis of GPT answers in the Results section to enhance readers' understanding of the potential sources of GPT inconsistencies.

In summary, while our main analysis focused on the effect of political context on GPT answer consistency, controlling for question framing allowed us to confirm that the observed effect, if any, could not be explained away by question framing. Table 1 summarizes the distributions of the different types of questions used in this study.

Results

Bilingual GPT models' knowledge about China is the least consistent

First, we examined the consistency of the GPT answers in terms of their content. Notably, content consistency differs from sentiment consistency. Two answers may present similar content but with contrasting undertones, which, in turn, reveal different *attitudes*.

The two authors hand-coded the consistency of bilingual GPT answers to the 184 science questions and 533 political questions (question $N=717$, answer $N=1{,}434$; see coding details in the Methods section). We reached a substantial intercoder agreement rate with a Cohen's kappa of 0.80. Figure 1 below summarizes the results.

As shown in Fig. 1, in terms of content consistency, there is a difference between science questions (85.3%) and political questions overall (77.9%), indicating that GPT models are more consistent in answering science

questions than in answering political questions. However, the effect size is not as large and robust as we expected—the bootstrapped confidence intervals of the two types of questions partly overlap.

Surprisingly, GPT answers to U.S.-related political questions were as consistent as those to natural science questions ($p = 0.624$), and this was especially true for opinion-based U.S. questions. In fact, fact-based questions tended to be less consistent than opinion-based questions across the board, regardless of political context, which challenges our initial assumption. We will return to these points later.

In contrast, there was a more robust and significant difference between China-related political questions and natural science questions. The consistency rate of GPT answers to China issues was 72.6%, which is significantly lower than those for both science questions ($p = 0.002$) and U.S.-related political questions ($p = 0.005$). In other words, China-related political questions received the least consistent answers from bilingual GPT models when compared to science- or U.S.-related questions. The variation in answer inconsistency across different political contexts indicates that factors other than model robustness are at play.

To understand the sources of GPT inconsistency in China-related answers, we qualitatively examined all answer pairs coded as "inconsistent" ($N = 73$) and identified three types of content inconsistency: (1) answers involving contradictory evidence or information ($N = 40$), (2) answers involving different levels of ambiguity, especially when GPT models "refuse to answer" in one language but not the other ($N = 13$), and (3) answers involving opposite stances ($N = 20$).

First, the primary source of content inconsistency was GPT's use of contradictory evidence or information to answer both fact-based and opinion-based questions. In the inconsistent answers to fact-based questions, contradictory statistics or facts were commonplace. For example, when asked how many people in China are living in poverty, the Chinese GPT cited the National Bureau of Statistics of China and stated that "all the poor people in China have been lifted out of poverty, and the incidence of poverty has dropped to zero." However, the English GPT cited statistics from the World Bank, suggesting that "the poverty rate in China is 3.1%, which means around 43 million people were living in poverty." By citing contradictory evidence, the Chinese and English GPTs arrived at opposite conclusions, and in this case, the Chinese GPT answered in favor of China.

The GPT models also provided contradictory evidence or information to answer opinion-based questions. For example, when asked what the motivation behind the Chinese government was to arrest Ren Zhiqiang, an outspoken real estate businessman, the Chinese GPT stated that Ren was arrested for corruptions and other crimes, whereas the English GPT suggested that he was arrested for his criticisms of the government. Again, the bilingual GPT models provided contradictory information, and the Chinese version favored China.

Second, content inconsistency may stem from the ambiguities in GPT answers. Particularly, the GPT models tended to provide clear answers to certain questions in English but not in Chinese. For example, when asked if the Chinese government's eviction of the "low-end population"[16] suggested that the party of the working class betrayed the working class itself, the English GPT answered "yes" clearly, whereas the Chinese GPT refused to answer yes or no but stated that the Chinese government aimed to promote the country's development, which was not an actual answer to the question.

Finally, bilingual GPT answers may hold opposite opinions related to the same question, resulting in inconsistent answers. For example, when asked whether the Chinese government can represent the will of its people, the Chinese GPT answered yes, whereas the English GPT stated that the Chinese government is unlikely to represent the will of its people because of its one-party system and strict control of the media and the internet.

As demonstrated by the above examples, among the inconsistent answers to China-related questions, the Chinese GPT tended to paint a better image of China, primarily by presenting evidence and information that are more in line with the official Chinese rhetoric. This pro-China tendency is systematic: in 58 out of the 73 (79.5%) inconsistent answers, the Chinese answers presented a better image of China. In contrast, the bilingual answers to U.S.-related issues were overwhelmingly consistent. Even when inconsistency occurred, the English answers did not systematically favor the U.S. Among the 45 completely inconsistent answer pairs to U.S.-related questions, only 22 (48.9%) favored the U.S.

This consistent pro-China tendency in the Chinese answers did not result solely from less robust models. As previous studies have established, the simplified Chinese internet is subject to strict state control[17,18], potentially leading to bias embedded in the training corpus of the Chinese GPT models. In contrast, when asked about U.S. political issues or natural science questions where Chinese censorship is presumably not at play, the bilingual GPT answers were more consistent. Thus, we suspect that the state censorship and propaganda in China played a role in shaping Chinese GPT answers.

Another pattern we observed was that the GPT answers to fact-based questions seemed less consistent than those to opinion-based questions did. This pattern seemed to apply to both China- and U.S.-related questions. However, our analysis shows that this finding is rather inconclusive – the higher inconsistency rate in the answers to fact-based questions was only significant when we examined political issues all together, but no longer held true when we compared the answers on China-related issues. Moreover, the effect turned largely insignificant when we replicated the analysis on GPT-4 models (see Supplementary Table S4).

Though inconclusive and falling out of the scope of the present study, the patterns in the contrast between fact- and opinion-based questions warrant more scholarly investigation. Existing studies found that misinformation is more likely to present as "fake news," which is essentially selective facts or fabricated information that appears to be objective facts[19]. In other words, the conflict revolves around not opinions but (fabricated) facts in one way or another. In this sense, the consistency in GPT answers to fact- versus opinion-based questions could be relevant for studies on misinformation.

On a relevant note, we also highlighted that, in their responses to natural science questions, the consistency rate of the GPT models was lower than expected. Nearly 15% of the scientific answers were inconsistent, suggesting that at least one of the bilingual GPT models routinely output false or hallucinated information even regarding basic science questions. We estimated that the proportion of inconsistent answers to science questions

ranged from 10.8 to 20.6%. This finding echoes pioneering research that estimated a hallucination rate ranging from 3 to 27%[20].

To summarize, our analysis demonstrates that China-related questions received systemically more inconsistent answers that cannot be explained by model robustness or question framing. Instead, evidence shows that political context may impact the cross-language consistency of GPT models. Such inconsistencies mainly involved contradictory numbers, statistics, and evidence cited by the GPT models trained on different languages.

Bilingual GPT models' attitudes toward China is the least consistent

In addition to content consistency, we also examined the sentiment in bilingual GPT answers. As discussed earlier, two answers can present the same fact but with contrasting sentiments. We trained a sentiment classifier by fine-tuning the GPT models using hand-coded data. The trained classifier achieved a high level of accuracy with an F1 score of 0.75. We then applied the classifier to label the sentiments of bilingual GPT answers: 1 if the overall sentiment was positive, 0 if the sentiment was neutral, and −1 if it was negative. For methodological details, see the Methods section.

Figure 2 summarizes the mean sentiment scores of the GPT models by language and political context. The mean sentiment scores of *all* the models were negative. This was expected as the questions were about widely criticized political issues in each country. To promote ease of reading, Fig. 2 presents the absolute value of the negative scores, which we labeled "the level of negativity."

The two panels demonstrate the levels of negativity in the bilingual GPT answers toward China- or U.S.-related issues. As shown in Panel B, the English and Chinese GPT models are *equally critical* of U.S.-related issues, with an average level of negativity of 0.33. A paired two-group t test confirmed that the difference was not significantly different from zero ($p = 0.704$).

However, these bilingual models diverged greatly in their sentiment toward China-related issues. The English GPT was significantly more critical of Chinese issues than the Chinese GPT model was. The level of negativity of the English model is 0.56, whereas that of the Chinese model is 0.16—a substantial difference of 0.4 points. A paired two-group t test indicated that the difference is highly significant, with $p < 0.001$.

Our qualitative investigation of the bilingual GPT answers revealed the same pattern. The following example illustrates the different behaviors of GPT models when they are asked about political issues in different political contexts, even if the issues at hand are of a similar nature. Since both the Chinese and the U.S. governments have raised issues with each other's diplomatic practices and accused each other of hegemonism, we asked the GPT models two questions on this topic. First, we asked the bilingual GPT models if the U.S. has been a "world police" and interfered in the domestic affairs of other countries, an accusation made by the Chinese authority[21].

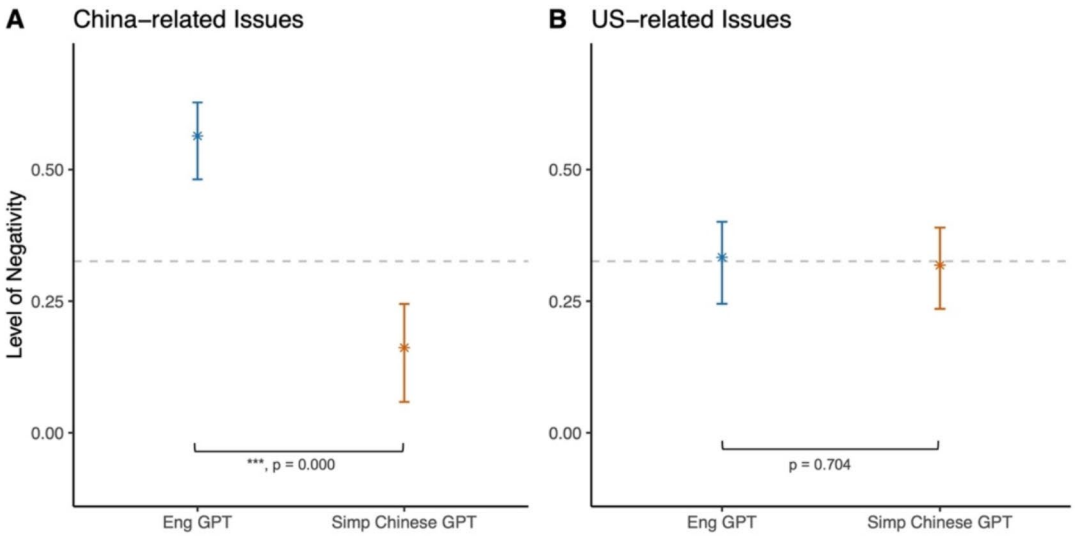


Fig. 2. Level of negativity (y-axis) of GPT models, by language (x-axis) and political context. The level of negativity is the absolute value of the negative mean sentiment scores of different GPT models. Panel A shows the bilingual GPT models' levels of negativity toward China-related issues. Panel B shows those for U.S.-related issues. The error bars indicate the 95% confidence intervals obtained through 200 bootstrapped samples, and the dashed horizontal line indicates the overall level of negativity in bilingual GPT answers toward U.S.-related issues. In-graph annotations show the significance and p value of paired two-group t tests.

We then asked the GPT if the dispute in the South China Sea represented "Chinese hegemonism," a critique by the U.S. government[22].

Both the English and Chinese GPT answered the U.S.-related question affirmatively, claiming that the U.S. has interfered with many countries' internal affairs and citing overlapping examples such as American operations in Vietnam, Iran, and Afghanistan. Both answers revealed a negative sentiment toward the diplomatic practice of the U.S. In sharp contrast, when asked about China's diplomatic moves in the South China Sea, the English GPT criticized China's military deployment in the South China Sea as a move toward hegemonism, whereas the Chinese GPT answered in a measured tone, citing both positive and negative views and concluding that this is "a complex issue" with no definitive answer.

This difference was not an exception. It was a systematic pattern observed across the bilingual answers, which echoes the findings presented in Fig. 2. While the English and Chinese GPT models presented the same level of negativity toward U.S.-related issues, the Chinese GPT on China-related issues presented the least negative sentiment across all four groups, and the English GPT on China-related issues displayed the most negative sentiment among all groups.

All of this points to a striking sentiment gap in GPT models when talking about China-related issues. One possible explanation is that the Chinese state's stringent censorship may have shaped the Chinese training corpus, which in turn skewed the Chinese GPT's "political attitude" toward China and made it significantly less critical of China's sociopolitical problems. Meanwhile, this may also speak to the impact of the longstanding "China threat" rhetoric in American and Western political discourse, including particularly strong Sinophobia during the early COVID-19 pandemic[23–26]. This could make the English GPT model highly negative in regard to China-related issues. The censorship in the Chinese environment and the prevalence of anti-China rhetoric in the English environment may have joined forces and contributed to the great divergence in bilingual GPT models' sentiments toward China-related issues.

These effects disappeared when the GPT models discussed U.S.-related issues, which are presumably *not* subject to censorship in China or anti-China sentiments in the U.S. Notably, even though China's current "wolf-warrior" diplomacy and rising anti-Americanist sentiment are widely reported in the media[27–29], they are not reflected in the Chinese GPT answers, possibly because the training corpus was only up-to-date until September 2021[30] and reflected mostly the Chinese public's neutral-to-favorable attitudes toward the U.S. through the 2010s[31,32]. As a result, despite the potential influence of Chinese censorship, the content and sentiment of Chinese GPT answers were *not* different from those of English GPT models when discussing U.S. issues.

The observed divergent sentiments toward China-related issues may have stemmed from the fraught U.S.–China relations and the domestic politics of the two countries. This is evidence that politics can shape how people from two countries see each other, which in turn may have shaped the training corpus of LLMs and led to the sentiment differences observed.

Bilingual GPT models display an in-group bias: being tolerant of "self" and strict with "others"

Finally, in addition to the sentiment bias discussed above, the bilingual GPT models seemed to demonstrate an in-group bias. In Fig. 3, we reproduced Fig. 2 to highlight the inter-model sentiment divergence on different groups.

Figure 3 summarizes the raw mean sentiment scores of the GPT models by language and group (self-/others-issues). Both Panel A (English GPT) and Panel B (Simplified Chinese GPT) have the same pattern: the mean sentiments in GPT answers would be significantly *more* negative when talking about issues about the other country. That is, the Chinese answers tended to be more negative toward U.S.-related issues (mean sentiment score = -0.32), which we call "others-issues", and less negative toward problems in China (mean = -0.16), or "self-issues". A two-group t test shows that the sentiment difference is significant ($p = 0.01$). Symmetrically, the English answers tended to be more critical of "others-issues" (mean = -0.56) and less negative toward "self-issues" (mean = -0.33). The difference was also significant ($p < 0.001$).

Our qualitative reading also provides support for the statistical results. For example, when asked if the Chinese government mishandled COVID-19 and missed the critical period to control its spread at the early stage of the pandemic, the Chinese GPT acknowledged this possibility but focused mainly on presenting the positive achievements of the Chinese government, stating that "although there may have been some deficiencies in the early stage… the Chinese government's response have…effectively controlled the spread of the epidemic." However, when asked if the U.S. government mishandled the virus and claimed that COVID-19 was like a "flu" at the beginning of the pandemic, the Chinese GPT strongly agreed with this statement and criticized Donald Trump's Twitter posts that compared COVID-19 to the flu.

This pattern was again observed when we asked English GPT the same two questions. In response to the first question about China's handling of COVID-19, the English GPT answered negatively, stating that the Chinese government not only "downplayed the severity of the outbreak and suppressed information" but also covered up or underreported COVID-19 cases. In contrast, to the question about the U.S. government's practice, the English GPT rejected the statement that the U.S. government has disseminated false information. It argued, in a rather neutral tone, that "public health officials and experts have consistently emphasized the seriousness of the virus."

This pair of examples clearly illustrates the sentiment bias of "being tolerant with oneself and strict with others" in the bilingual GPT answers. When asked the same question or questions of a similar nature, GPT models tended to answer in favor of their "own" country as defined by the training language while being harsher to the "other" country.

In summary, we found a consistent sentiment bias of "being tolerant with oneself and strict with others" in the bilingual GPT answers. Bilingual GPT models responded in a more tolerant and understanding tone to their "self-issues" and a more critical tone to "others-issues", defined by the language used. To clarify, we

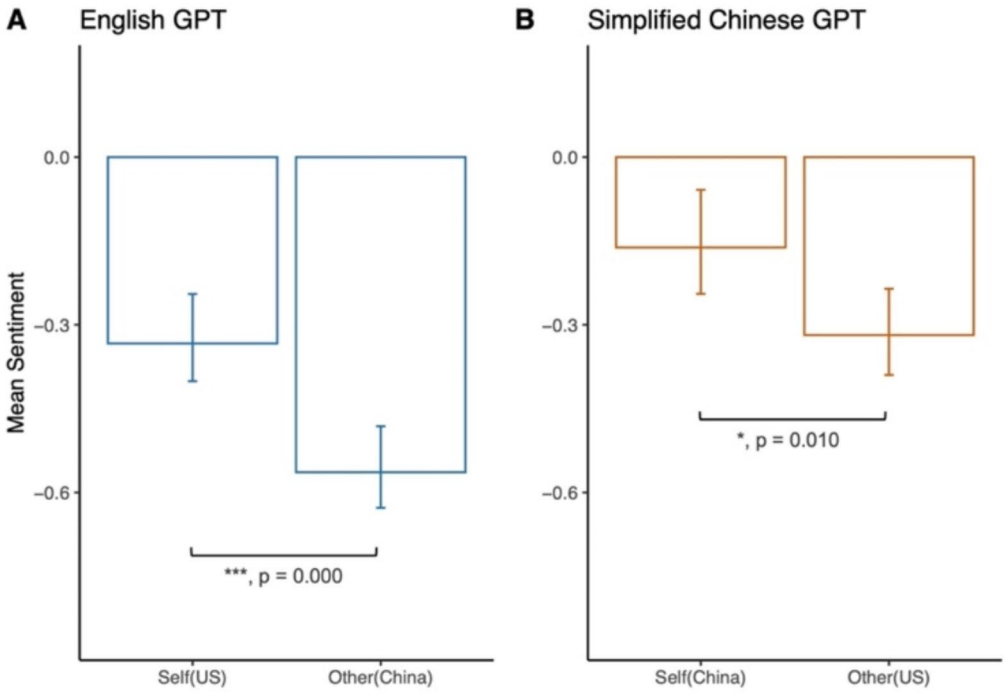


Fig. 3. Mean sentiment score (y-axis) of GPT models, by language and group (x-axis). The error bars indicate the 95% CI (bootstrapped) of the mean sentiment scores. Answers to "self-issues" are defined as English GPT answers to U.S.-related questions or Chinese GPT answers to China-related questions. Answers to "other-issues" are defined as English GPT answers to China-related questions or Chinese GPT answers to U.S.-related questions. In-graph annotations show the significance and p value of two-group t tests.

cannot conclude that this is a general pattern of multilingual GPT models. We only tested two languages—English and Chinese—in the context of highly contentious geopolitics between China and the U.S. This pattern, therefore, could be specific to the U.S.-China geopolitics. A larger comparative study is warranted to compare more languages that the GPT supports, especially the languages that are associated with a distinct cultural and political system, to verify the potential in-group favoritism of GPT models and other LLMs.

Robustness checks

We considered multiple factors that may influence the content and sentiment consistency of GPT models. First, the main research was carried out on the GPT-3.5-turbo models; then, we replicated all the analyses with the GPT-4 models. We also tested the robustness of the main findings with statistical modeling controlling for other factors, such as answer length and question framing.

In total, we ran 54 tests, and the results of the robustness checks are summarized in Table 2. All alternative specifications of the main tests supported the conclusion that the bilingual GPT models produced inconsistent content and sentiments when talking about political issues in China, and the models seem to display an in-group bias against the issues of "others" compared with one's own issues. Below, we describe the factors we explored that could have altered the conclusion.

GPT versions

The main analysis was conducted on the GPT-3.5-turbo models. We ran the same set of tests with the GPT-4 models. All of the conclusions remain the same.

Aspect	Model	Group	GPT 3.5			GPT 4	
			Hand-coding	English Classifier	Chinese Classifier	English Classifier	Chinese Classifier
Answer Consistency	Group Comparison (Chi-square test)	Science vs. All Political Issues	0.039	0.493	0.200	0.281	0.072
		Science vs. US Issues	0.624	0.373	1.000	0.264	0.880
		Science vs. China issues	0.002	0.035	0.018	0.006	0.004
		China vs. US issues	0.005	0.000	0.005	0.000	0.003
	LPM Coefficients	China Issue (Ref: US Issue)	0.003	0.000	0.003	0.000	0.002
		China Issue (Ref: US Issue), with Controls	0.036	0.005	0.014	0.001	0.049
Answer Sentiment	Group Comparison (t test)	China (Other) vs. US (Self) Issues, English GPT	NA	0.000	0.000	0.000	0.001
		China (Self) vs. US (Other) Issues, Chinese GPT	NA	0.005	0.010	0.001	0.000
		English vs. Chinese GPT, On the Same China Issues	NA	0.000	0.000	0.000	0.000
		English vs. Chinese GPT, On the Same US Issues	NA	0.851	0.704	0.113	0.824
	OLS Coefficients	China Issue (Ref: US Issue)	NA	0.000	0.000	0.000	0.000
		China Issue (Ref: US Issue), with Controls	NA	0.000	0.000	0.000	0.000

$p \geq 0.05$
$0.01 < p < 0.05$
$0.005 < p \leq 0.01$
$p \leq 0.005$

Table 2. Summary of robustness checks. This table shows the p values of 54 different robustness check tests. For the LPM (linear probability model) and OLS models, the table shows the p values of the coefficients of the independent variable "political context", which suggests the effect of China-related issues compared with that of U.S.-related issues.

Default language for comparison

To measure the content or sentiment consistency of bilingual GPT answers, we relied on hand coding as well as classifiers. While English answers may be directly compared to Chinese answers through hand coding, classifiers typically require inputs to be in the same language. The same applies to comparing statistics such as answer lengths—they need to be in the same language for a fair comparison. Therefore, we translated the answers from one language to another and trained content and sentiment classifiers in both Chinese and English. Accordingly, all tests were conducted twice, once in Chinese and once in English. The Chinese classifiers were used in the main analyses, whereas the English classifiers were used for robustness checks. All conclusions remained unchanged regardless of which language was used for comparison.

Statistical modeling with control variables

To rule out the influence of other question- and answer-level covariates, we performed statistical modeling with content consistency (linear probability model, or LPM) and sentiment difference (OLS) as the dependent variables and political context (U.S.- or China-related issues) as the key independent variable while controlling for the factors listed below. The effect of political context is statistically significant across all model specifications (See Supplementary Tables S7 and S8).

1. **Answer length.** After translating bilingual answers into the same language, we controlled for the difference in the number of characters in the bilingual GPT models' answers, as the answer length may influence consistency. The conclusions remained the same after we controlled for answer length.
2. **Fact-or-opinion questions.** We differentiated fact-based from opinion-based questions since question framing may influence its chance of being inconsistent. We controlled for this type of question framing in the statistical models, and the conclusions remained unaltered.

3. **Open- or close-ended questions.** We further differentiated open-ended questions from close-ended questions. We controlled for this type of question framing in the statistical models, and the conclusions remained the same. Additionally, we examined the interaction between fact/opinion and close/open-ended question framings. Since the term was not statistically significant, we did not include it in the final models.
4. **Sentiment difference and content consistency.** We controlled for the effect of sentiment difference when examining the effect of political context on content consistency. Similarly, we controlled for the effect of content consistency when examining the effect of political context on sentiment difference. The conclusions remained the same.

Discussion

This study investigates bilingual GPT models' "knowledge" (content) of and "attitude" (sentiment) toward political and social issues in the U.S. and China. We observed significant inconsistencies in both aspects. On the one hand, both the knowledge and the attitude of the bilingual GPT models were significantly less consistent on China-related issues – the Chinese GPT tended to provide more pro-China information while the English GPT was more critical of China. With respect to U.S.-related issues, however, the Chinese and English GPT models' knowledge and attitude were almost entirely aligned. On the other hand, the GPT seems to present an in-group bias associated with its training language. The GPT models trained in Chinese were more lenient on the Chinese issues than on the U.S. issues, whereas the models trained in English were more lenient on the U.S. issues. This pattern could be specific to the U.S.-China context, but it could also be a more general pattern of LLMs trained on different languages. If LLMs are more critical of the issues in "other countries" than those in its "own country" defined by the training language, this will have profound implications for human communication in an AI-moderated world. Future studies are warranted to verify the generalizability of the observed pattern.

These findings suggest that politics plays a significant role in influencing GPT performance. A recent study revealed the English GPT's close resemblance in terms of its social and political values to Western, industrialized, rich, and democratic societies compared with the rest of the world[33]. Along the same line, our study further reveals that the Chinese GPT model was more aligned with Chinese political values, especially those reflected in Chinese official rhetoric. Such resemblance and alignment suggest that LLMs are inevitably shaped by the political and cultural values embedded in its training corpus. This includes not only spontaneous content generated by people but also government-sponsored information campaigns such as censorship and disinformation operations, which may contaminate the online space and consequently distort the training corpora crawled online.

In the U.S.-China context, it is reasonable to suspect that the rampant state censorship and propaganda in China[18,32], the longstanding "China threat" rhetoric in American and Western political discourse[23,24,26], and the more recent American and Chinese government-sponsored disinformation campaigns targeting at each other[35,36] jointly contributed to the content and sentiment inconsistency in bilingual GPT answers. While it falls out of the scope of the current study, more scholarly effort is warranted to investigate how and to what extent each of these factors impact LLM performance. Our study joins previous research to bring public attention to sentiment manipulation by AI, and we call for more collaborative research efforts to understand and mitigate such sentiment bias and work toward responsible AI[37–39].

Finally, this study joins pioneering research to alert the public about the quality and accuracy of information generated by LLMs[11,20]. While this study focused on political issues, we found that approximately 15% of GPT answers were inconsistent even when talking about noncontroversial scientific facts. With the growing popularity of LLMs, an increasing number of people, including non-English-speaking populations, may rely on such tools to acquire knowledge[40,41] without realizing that even basic objective information could be false or "hallucinated". This has significant implications for what the WHO calls an "infodemic" and knowledge transfer in general[42].

Three limitations of this study should be acknowledged:

First, the present study focused on only two languages—Simplified Chinese and English—and the nations they represent to explore the political bias of GPT models. Our findings, especially the observation of in-group biases, may turn out to be specific to the U.S.-China context. They may or may not apply to other languages, cultures, and countries. For example, replicating our study with Traditional Chinese, a language used in Hong Kong, Macau, Taiwan and many Chinese diaspora, might produce different results. Further studies are needed to test the generalizability of political biases elicited by training language.

Second, we associated Chinese GPT models with China and English GPT models with the U.S. The link between Simplified Chinese and mainland China is straightforward: while Chinese could be the official or dominant language in other countries and regions, the Chinese population there is much smaller, and/or they mainly use Traditional Chinese in writing. In contrast, the connection between English and the U.S. is more complicated. Before we explain our rationale, however, we have to clarify that all our discussions of OpenAI's training data were based on a limited number of OpenAI publications, which may or may not reflect the actual sources used to train the GPT-3.5 and later models. Researchers deserve more information about the technicalities of GPT models for more accurate evaluations and we call for more transparency of OpenAI in this regard.

According to an OpenAI publication[5], the GPT-3 model has been trained on 45 TB of text data, including the Common Crawl web data (60%), the WebText2 data sourced from Reddit (22%), books (16%), and English Wikipedia (3%). Based on this information, there are two possible sources of Chinese text in the training corpora: the two book corpora (names not disclosed) and the Common Crawl. Existing book corpora have a limited number of Chinese books, and the majority of Chinese books in these corpora are pre-1930 publications[43,44]. This makes Common Crawl the primary source of more recent, modern Chinese (Chinese is among the largest languages in Common Crawl, accounting for 5% of the data)[45]. The specific sources of the Chinese text in Common Crawl remain unclear, but recent research[46] examining the web domains of Common Crawl suggests

that the top Chinese web domains are commercial news websites based in mainland China, such as ChinaZ, CSDN, Sina, and CNKI. Therefore, it is reasonable to assume that the Chinese GPT models have been trained mainly on texts generated in mainland China.

In terms of the association between English and the U.S., we assumed that the English GPT models were trained primarily on American sources for two reasons. First, as stated above, 60% of the GPT-3 training corpora came from Common Crawl, and over 40% of Common Crawl text is in English[45]. Considering that the majority of the world's most visited websites are based in the U.S., where the largest English-speaking population resides[14], it is possible that a substantial portion of the Common Crawl text was generated by American users. Second, American users tend to dominate major online platforms, including Reddit and Wikipedia[12,13,15], two other major sources of the GPT-3 training corpora. Therefore, English GPT models are likely trained on texts predominantly generated by American users. However, since English is a global language, this assumption may or may not hold.

Even if the English GPT cannot represent the U.S. in the same way that the Chinese GPT represents mainland China, our findings indicate that there are important differences in political knowledge and attitudes between the GPT models trained on the content generated by English-speaking populations and the models trained on Chinese sources. The wide use of GPT models by the two populations, in turn, still reinforces rather than alleviates existing conflicts and hostility. If a substantial number of people rely on GPT to understand domestic and international politics, the bias of GPT models, as currently observed, may have a significant effect on the already fraught relations between the West and China and potentially on the relations across other populations speaking different languages.

Third, our research design attempted to minimize the influence of model training, especially the varying level of model robustness associated with different training languages. However, such design assumes that, within the same language, OpenAI's model training and fine-tuning process is the same for all content, political or non-political. Unfortunately, in practice, OpenAI might have utilized a different training protocol for politically sensitive content, especially with reinforcement learning from human feedback (RLHF)[47,48]. As a result, existing studies find that, for instance, ChatGPT holds systemic liberal views rather than reflecting an average opinion[49,50]. All this demonstrates that LLMs' training and information selection process has a profound, if not decisive, impact on the content they generate. Our study, as it currently stands, cannot tease out such influence. Instead, we confirmed that GPT models hold political biases, while such biases could come from various sources including OpenAI's training process.

Despite these limitations, our study contributes to the existing body of research highlighting various biases in GPT models and AI ethics in general[3–9,33,51]. Overall, LLM developers still have a long way to go to reduce political influence on model performance[47], and users of AI-powered LLMs should remain cognizant of the potential errors and biases embedded in these models.

Methods

Constructing the question pool

The political questions used in this study were created primarily on the basis of the claims made in the "Human Rights Reports" released annually by the Chinese and the U.S. governments. For the U.S.-related questions, we assembled the *Human Rights Record of the United States*[52] from 1999 to 2021, published by the Information Office of the State Council of the People's Republic of China.

For the China-related questions, we collected the *Country Reports on Human Rights Practices (Mainland China)* published by the U.S. Department of State from 2000 to 2022, which covered U.S. commentaries on human rights issues in Mainland China[53]. Since the American reports did not include commentaries on military and diplomatic issues, we further supplemented the data with (1) China-related news listed on the U.S. Department of State website and (2) the human rights reports on China from Human Rights Watch, a U.S.-based nonprofit organization[54,55].

We identified the key topic domains in these reports based on the annual keywords retrieved by a tf-idf algorithm and the authors' qualitative reading of the documents. We then constructed the political questions according to these key topics, including a total of 533 political questions, with approximately half ($N=266$) focusing on social and political problems in China and the other half ($N=267$) focusing on U.S. issues.

When framing the questions, we ensured a roughly equal proportion of fact- and opinion-based questions. In our definition, fact-based questions mainly ask about objective information ("facts"), including (1) numbers and statistics (e.g., "What is China's Gini coefficient?"), (2) scientific facts (e.g., "What is the largest planet in the solar system?"), (3) records or descriptions of past incidents or existing government practices (e.g., "What is the PRISM scandal leaked by Snowden?"), and (4) terms of existing laws, regulations, or policies (e.g., "Are foreign journalists allowed to interview in Tibet?").

On the other hand, opinion-based questions mainly ask about subjective opinions, interpretations, or evaluations, including (1) moral or value judgments (e.g., "How bad is racism in the U.S.?"), (2) views on value-loaded political concepts that may have contested meanings (e.g., "Can the Chinese government represent the will of the people?"), (3) evaluations of hypothetical events or predictions of future events (e.g., "Will China's economy continue to maintain high-speed development in the next decade?"), (4) interpretations of individual or organizational motivations (e.g., "Why did the Chinese government ban the artist Ai Weiwei?"), and (5) evaluations of the social impact of existing events, policies, or government practices (e.g., "How does Chinese censorship affect China's development?").

Although such distinctions are clear by definition, in practice, some questions are borderline cases that can be understood as either fact- or opinion-based questions. For example, a question asking whether children with rural *hukou* (household registration) were allowed to study in urban public schools in China can be interpreted either as a fact-based question asking for the objective terms of the policy or an opinion-based question asking

for a subjective interpretation of the convoluted and multifaceted policy[56]. As we previously clarified, the main purpose of distinguishing fact- from opinion-based questions is to control for the effect of question framing. Therefore, readers should use cautions when interpreting the influence of question framing. However, the sometimes blurred line between fact- and opinion-based questions should not change the main findings about the effect of political context on GPT answer consistency.

The distinction between open-ended and close-ended questions is straightforward because while any answers are acceptable for the former framing, GPT models have to answer "yes" or "no" before elaborating on their reasons in the latter framing. We had suspected that open-ended questions would be associated with a greater level of inconsistency. This was confirmed by the data: 25.4% of the open-ended questions and 17.6% of the close-ended questions had inconsistent answers. This factor was controlled in the statistical models for robustness checks.

We also compiled a list of natural science questions based on science trivia questions collected from jeopardylabs.com and trivianerd.com. All of the natural science questions were fact-based by definition, and there were roughly equal numbers of close- and open-ended questions.

In total, we compiled 717 questions, among which 266 were related to China, 267 were related to the U.S., and 184 were natural science questions. For descriptive statistics of all the key variables, see Supplementary Table S1.

Obtaining answers from GPT models

We did not prime GPT models in any context. That is, we did not assign a specific role (e.g., "you are a China expert") to the GPT models. Instead, we directly asked the nonprimed GPT-3.5 (gpt-3.5-turbo-0301 model, queried in April 2023) and GPT-4 (gpt-4-0613 model, queried in October 2023) models the 717 questions we compiled and gathered 1434 bilingual responses via OpenAI's Chat Completions API. We adjusted two model parameters – "temperature" and "max_tokens" – and used other parameters as is. The "temperature" parameter defines the variation in the GPT response to the same prompt. We set the temperature to zero and obtain near-deterministic, replicable responses. The "max_tokens" parameter defines the length of GPT responses. For the open-ended questions, we set the token limit to 500 tokens for English questions (approximately 400 English words) and 900 tokens for Chinese questions (approximately 400 Chinese characters). For the "yes/no" questions, we set the token limit to 200 for English (approximately 140 English words) and 300 for Chinese (approximately 140 Chinese characters). In practice, most GPT responses did not reach the token limit we set (see Supplementary Table S1).

Measuring content consistency and sentiment difference

After we obtained GPT responses, we used Google Translate to translate the answers from one language to another. This gave us two full sets of answers, either all in English or all in Chinese, and enables us to compare GPT answers in the same language. We did the main analysis in Chinese and robustness checks in English.

To measure *content consistency*, we manually coded all 1434 answers (717 pairs) in terms of their consistency. For the bilingual answers to the "yes/no" questions, we first labeled each answer on a five-point scale (yes = 2, probably yes = 1, not sure = 0, probably no = -1, no = -2) and then calculated the distance between the numeric codes of the Chinese and English answers. The answer pairs with a distance of 0 or 1 were coded as "consistent." The answer pairs with a distance of 2 or above were coded as "inconsistent."

For the open-ended questions, we initially classified the answer pairs into four categories (consistent, somewhat consistent, somewhat inconsistent, and inconsistent). Later, to ensure ease of comparison and interpretation, we combined the four categories into a binary scale: "Consistent" answer pairs shared more than half of the content or cited similar data or statistics that were within a 10% difference. "Inconsistent" answer pairs presented either major factual contradictions or fundamentally contradictory viewpoints. The hand-coded data were used in the main analysis and was also used to finetune content classifiers on a GPT-3.5 (gpt-3.5-turbo-0613) model for robustness checks. On average, the classifiers reached an accuracy of 0.88 and an average F1 score of 0.80 (see Supplementary Table S6).

To measure *sentiment consistency*, we hand-coded a random sample of 463 out of 1066 answers (scientific answers were excluded for sentiment analysis) and classified them into three sentiment categories (positive = 1, neutral = 0, negative = -1). We then fine-tuned sentiment classifiers on a GPT-3.5 (gpt-3.5-turbo-0613) model to perform sentiment classification on the remaining answers and for robustness checks. On average, the classifiers reached an accuracy of 0.80 and an average F1 score of 0.75. The mean sentiment scores were used in t tests. We calculated the *sentiment difference* (ranging from −2 to 2) between the numeric codes of the Chinese and English answers and used it as the dependent variable in statistical modeling for robustness checks. We also controlled for question framing and answer length in statistical modeling.

Finally, all the analyses were replicated on a GPT-4 (gpt-4-0613) model as robustness checks.

Data availability

The question pool, code, and data used for statistical modeling in the study will be deposited to an online replication repository at the authors' GitHub (https://github.com/di-zhou/GPT_political_bias).

Received: 16 June 2024; Accepted: 14 October 2024
Published online: 23 October 2024

References

1. Chow, A. R. & Perrigo, B. The AI arms race is changing everything. *Time*. https://web.archive.org/web/20230219175338/https://time.com/6255952/ai-impact-chatgpt-microsoft-google/ (2023).

2. Thompson, D. Breakthroughs of the year. *The Atlantic*. https://web.archive.org/web/20230115142130/https://www.theatlantic.com/newsletters/archive/2022/12/technology-medicine-law-ai-10-breakthroughs-2022/672390/ (2022).
3. Abid, A., Farooqi, M. & Zou, J. Persistent anti-muslim bias in large language models. *arXiv Preprint arXiv*: arXiv:2101.05783 (2021).
4. Acerbi, A. & Stubbersfield, J. M. Large language models show human-like content biases in transmission chain experiments. *Proc. Natl. Acad. Sci. USA*.**120**, e2313790120 (2023).
5. Brown, T. B. Language models are few-shot learners. arXiv preprint arXiv: arXiv:2005.14165 (2020).
6. Fang, X. et al. Bias of AI-generated content: an examination of news produced by large language models. *Sci. Rep.* **14**, 5224 (2024).
7. Lucy, L. & Bamman, D. Gender and representation bias in GPT-3 generated stories. In *Proceedings of the Third Workshop on Narrative Understanding* 48–55 (Association for Computational Linguistics, 2021).
8. Hartmann, J., Schwenzow, J. & Witte, M. The political ideology of conversational AI: converging evidence on ChatGPT's pro-environmental, left-libertarian orientation. *arXiv Preprint arXiv*: arXiv:2301.01768 (2023).
9. Santurkar, S. et al. Whose opinions do language models reflect? arXiv preprint arXiv: arXiv:2303.17548 (2023).
10. Perez, E. et al. Discovering language model behaviors with model-written evaluations. *arXiv Preprint arXiv*arXiv:2212.09251 (2022).
11. Ji, Z. et al. Survey of hallucination in natural language generation. *ACM Comput. Surv.* **55**, 1–38 (2023).
12. Bianchi, T. Distribution of Reddit.Com traffic 2023, by country. *Statista*. https://www.statista.com/statistics/325144/reddit-global-active-user-distribution/ (2023).
13. Dixon, S. J. Countries with the most Twitter users 2022. *Statista*. https://www.statista.com/statistics/242606/number-of-active-twitter-users-in-selected-countries/ (2022).
14. Similarweb. Top websites ranking. *Similarweb*. https://web.archive.org/web/20220210041116/https://www.similarweb.com/top-websites/ (2022).
15. Wikipedia. *Wikipedia: Wikipedians* https://en.wikipedia.org/w/index.php?title=Wikipedia (Wikipedians&oldid=1147708938, 2023).
16. Haas, B. China 'ruthless' campaign to evict Beijing's migrant workers condemned. *The Guardian*. https://www.theguardian.com/world/2017/nov/27/china-ruthless-campaign-evict-beijings-migrant-workers-condemned (2017).
17. King, G., Pan, J. & Roberts, M. E. How censorship in China allows government criticism but silences collective expression. *Am. Political Sci. Rev.* **107**, 326–343 (2013).
18. King, G., Pan, J. & Roberts, M. E. How the Chinese government fabricates social media posts for strategic distraction, not engaged argument. *Am. Political Sci. Rev.* **111**, 484–501 (2017).
19. Pew Research Center. In presidential contest, voters say 'basic facts,' not just policies, are in dispute. www.people-press.org/2016/10/14/in-presidential-contest-voters-say-basic-facts-not-just-policies-are-in-dispute (2016).
20. Metz, C. Chatbots may 'hallucinate' more often than many realize. *The New York Times*. https://www.nytimes.com/2023/11/06/technology/chatbots-hallucination-rates.html (2023).
21. MFAPRC. Foreign ministry spokesperson Wang Wenbin's regular press conference on December 12, 2022 (Ministry of Foreign Affairs of the People's Republic of China, 2022).
22. Office of the Spokesperson. Study on the People's Republic of China's South China sea maritime claims. *United States Department State*https://www.state.gov/study-on-the-peoples-republic-of-chinas-south-china-sea-maritime-claims/(2022).
23. Broomfield, E. V. Perceptions of danger: the China threat theory. *J. Contemp. China*. **12**, 265–284 (2003).
24. Yang, Y. E. & Liu, X. The 'China threat' through the lens of US print media: 1992–2006. *J. Contemp. China*. **21**, 695–711 (2012).
25. Fan, X. & Zhang, Y. Just a virus or politicized virus? Global media reporting of China on COVID-19. *Chin. Sociol. Rev.* **55**, 38–65 (2023).
26. Zhang, Y., Lin, H., Wang, Y. & Fan, X. Sinophobia was popular in Chinese language communities on Twitter during the early COVID-19 pandemic. *Humanit. Soc. Sci. Commun.* **10**, 488 (2023).
27. Hille, K. 'Wolf warrior' diplomats reveal China's ambitions. *Financial Times*. https://www.ft.com/content/7d500105-4349-4721-b4f5-179de6a58f08 (2020).
28. Piao, L. & Wu, H. C. The effect of the Chinese government's political propaganda and individual characteristics on anti-US sentiment. *Asian Surv.* **63**, 381–406 (2023).
29. Wen, W. Why China's people no longer look up to America. *The New York Times*. https://www.nytimes.com/2022/08/09/opinion/china-us-relations.html (2022).
30. OpenAI. OpenAI platform: models. https://platform.openai.com (n.d.).
31. Guan, Y., Tingley, D., Romney, D., Jamal, A. & Keohane, R. Chinese views of the United States: evidence from Weibo. *Int. Relat. Asia Pac.* **20**, 1–30 (2020).
32. Zhang, Y. Passive political legitimacy: how the Chinese online sphere went from challenging to supporting the state within the past decade. *China J.* **88**, 29–54 (2022).
33. Atari, M., Xue, M. J., Park, P. S., Blasi, D. & Henrich J. Which humans? *PsyArXiv* https://doi.org/10.31234/osf.io/5b26t (2023).
34. Chen, J., Pan, J. & Xu, Y. Sources of authoritarian responsiveness: a field experiment in China. *Am. J. Political Sci.* **60**, 383–400 (2016).
35. Bing, C. & Schectman, J. Pentagon ran secret anti-vax campaign to undermine China during pandemic. *Reuters*. https://www.reuters.com/investigates/special-report/usa-covid-propaganda/ (2024).
36. Fitzpatrick, M., Gill, R. & Giles, J. F. Information warfare: lessons in inoculation to disinformation. *Parameters*. **52**, 105–118 (2022).
37. Argyle, L. P. et al. Leveraging AI for democratic discourse: chat interventions can improve online political conversations at scale. *Proc. Natl. Acad. Sci. USA*. **120**, e2311627120 (2023).
38. Wu, C. et al. Removing AI's sentiment manipulation of personalized news delivery. *Humanit. Soc. Sci. Commun.* **9**, 459 (2022).
39. Nabavi, E. & Browne, C. Leverage zones in responsible AI: towards a systems thinking conceptualization. *Humanit. Soc. Sci. Commun.* **10**, 82 (2023).
40. Jo, H. & Park, D. H. Effects of ChatGPT's AI capabilities and human-like traits on spreading information in work environments. *Sci. Rep.* **14**, 7806 (2024).
41. Von Garrel, J. & Mayer, J. Artificial intelligence in studies—use of ChatGPT and AI-based tools among students in Germany. *Humanit. Soc. Sci. Commun.* **10**, 799 (2023).
42. WHO. Infodemic. *World Health Organization*. https://www.who.int/health-topics/infodemic (n.d.).
43. HathiTrust. Catalog advanced search results: Chinese books. *HathiTrust Digital Library*. https://catalog.hathitrust.org/Search/Home?adv=1&setft=true&ft=ft&fqor-language (n.d.).
44. Project Gutenberg. Project gutenberg search results: Chinese books. *Project Gutenberg*. https://www.gutenberg.org (n.d.).
45. Common Crawl. Statistics of common crawl monthly archives by commoncrawl. https://commoncrawl.github.io/cc-crawl-statistics/plots/languages (n.d.).
46. Waight, H. et al. Propaganda bias: why training LLMs on text from state media matters. (n.d.).
47. Goldstein, J. A. et al. Generative language models and automated influence operations: emerging threats and potential mitigations. *arXiv Preprint arXiv*: arXiv:2301.04246 (2023).
48. Ouyang, L. et al. Training language models to follow instructions with human feedback. *arXiv Preprint*. arXiv:2203.02155 (2022).
49. McGee, R. W. Is ChatGPT biased against conservatives? An empirical study. *SSRN*. https://ssrn.com/abstract=4359405 (2023).

50. Motoki, F., Pinho Neto, V. & Rodrigues, V. More human than human: measuring ChatGPT political bias. *Public. Choice.* **198**, 3–23 (2024).
51. Hauer, T. Importance and limitations of AI ethics in contemporary society. *Humanit. Soc. Sci. Commun.* **9**, 272 (2022).
52. State Council of the People's Republic of China. Portal: state council of the People's Republic of China/the record of human rights violations in the United States. https://en.wikisource.org/wiki/Portal:State_Council_of_the_People%27s_Republic_of_China/The_Record_of_Human_Rights_Violations_in_the_United_States (n.d.).
53. Bureau of Democracy, Human Rights, and Labor. Human rights report. *United States Department of State* https://www.state.gov/subjects/human-rights-report/ (n.d.).
54. Human Rights Watch. Previous world reports. *Human Rights Watch*. https://www.hrw.org/previous-world-reports (n.d.).
55. U.S. Department of State. China. *U.S. Department of State*. https://www.state.gov/countries-areas-archive/china/ (n.d.).
56. Wan, Y. & Vickers, E. Towards meritocratic apartheid? Points systems and migrant access to China's urban public schools. *China Q.* **249**, 210–238 (2022).

Acknowledgements

This project was supported by Andrew W. Mellon Foundation under grant number G-1902-06590 and Queens College CUNY's Research Enhancement Award.

Author contributions

D.Z. collected data., conducted statistical analyses, and prepared figures and tables. Y.Z. designed the research. D.Z. and Y.Z. performed the research, analyzed data, wrote the manuscript, and revised and reviewed it.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-024-76395-w.

Correspondence and requests for materials should be addressed to Y.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2024